Department of Computer Science
Technical University of Munich

Esolution
Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Operations Research

Exam:	IN0024 / Endterm	**Date:**	Wednesday 6th August, 2025
Examiner:	Prof. Dr. Martin Bichler	**Time:**	17:00 – 19:00

	P 1	P 2	P 3	P 4	P 5	P 6	P 7
I							
II							

Working instructions

- This exam consists of **24 pages** with a total of **7 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 120 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **non-programmable pocket calculator**
  - one **analog dictionary** English ↔ native language
  - one **ruler/geo triangle**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Answer the exam questions in English or German.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from ______ to ______ / Early submission at ______

Problem 1 **Multiple Choice Questions (14 credits)**

Mark all correct answers (there can be more than one):

Scoring: Each question is worth up to 2 points. You receive 2 points if you select all correct options and no incorrect ones. Selecting one incorrect option or not selecting a correct option reduces the score to 1 point. Any greater deviation or selecting no option results in 0 points.

Mark correct answers with a cross	☒
To undo a cross, completely fill out the answer option	■
To re-mark an option, use a human-readable marking	×■

a) Which of the following statements are correct?

- ☒ Every linear program has a dual program.
- ☒ If the primal problem is unbounded, then the dual problem is infeasible.
- ☒ If a primal constraint is not tight, then its corresponding dual variable is zero.
- ☐ If the primal problem is infeasible, then the dual problem is unbounded.

b) Which of the following statements about Christofides' algorithm (as presented in the lecture) are correct?

- ☒ In the execution of the algorithm a Hamiltonian cycle is computed.
- ☒ In the execution of the algorithm an Eulerian tour is computed.
- ☒ The solution produced by the algorithm is at most twice as costly as any optimal solution.
- ☒ The algorithm applies only to metric TSP instances.

c) Which of the following statements about Cutting Planes are correct?

- ☒ An ideal outcome of cutting planes is the convex hull of all integer feasible solutions.
- ☐ The cutting plane method always ends with two different Gomory cuts.
- ☒ Combining cutting planes with Branch-and-Bound can speed up solving integer programs.
- ☐ Cutting planes remove integer solutions from the feasible region of the LP relaxation.

d) Suppose you've solved an LP in standard form and found an optimal basis. Now you perturb the right-hand side of a constraint by Δ. Which of the following sensitivity statements are always true (assuming the basis stays optimal)?

- ☐ The shadow price vector π can be used to find the change in the values of the basic variables.
- ☐ The shadow price vector π should be updated.
- ☒ The solution can become degenerate for some Δ values.
- ☒ The shadow price vector π can be used to find the change in the value of the objective function.

e) Let E be a finite set. For which of the following collections $I \subseteq \{S \mid S \subseteq E\}$ is (E, I) a matroid?

- ☐ $I = \{S \subseteq E : |S| \text{ is odd}\}$.
- ☒ Let the ground set E be the edge set of an undirected graph $G = (V, E)$. Define $I = \{S \subseteq E : S \text{ contains no cycle in } G\}$.
- ☒ $I = \{S \mid S \subseteq E\}$.
- ☐ $I = \{S \subseteq E : |S| \text{ is even}\}$.

f) Let $A \in \mathbb{R}^{m \times n}$ be a totally unimodular matrix and $b \in \mathbb{Z}^m$. Investigate the linear program $\max\{c^T x : Ax \leq b\}$. Which claims are correct?

- [x] The determinant of every submatrix of A has values in $\{+1, -1, 0\}$.
- [x] If c is integral, every optimal vertex solution x^* has integral value $c^T x^*$.
- [ ] There exists a bipartite graph G such that A is the incidence matrix of G.
- [ ] Adapting A and b such that the constraint $x \geq 0$ is added can destroy total unimodularity of A.

g) Which of the following are true?

- [ ] For a constrained optimization problem, a KKT point is always a local optimum.
- [ ] Gradient methods with momentum always need less iterations to converge than gradient methods without momentum.
- [ ] Gradient Descent methods always converge to the global minimum for strictly convex functions, independent of the chosen step-sizes.
- [x] For a constrained convex optimization problem a KKT point is always a global minimizer, if Slaters condition holds.

Additional space for notes:

Problem 2 Sensitivity (16 credits)

A café produces two free drinks: *Lemonade* (x_1) and *Iced Tea* (x_2). Each batch of:

- Lemonade uses 4 L of water, 1 kg of sugar, and 1 kg of fresh lemon;
- Iced Tea uses 2 L of water, 2 kg of sugar, and 0.5 kg of tea leaves.

Available per day: 100 L water, 40 kg sugar, 30 kg lemons, 20 kg tea leaves.
The café wants to find a daily production plan that produces both drinks, so they solve the following feasibility problem:

$$\begin{aligned} \max\ & 0x_1 + 0x_2 & \\ \text{s.t. } & 4x_1 + 2x_2 \leq 100 & \text{(water)} \\ & 1x_1 + 2x_2 \leq 40 & \text{(sugar)} \\ & 1x_1 \leq 30 & \text{(lemons)} \\ & 0.5x_2 \leq 20 & \text{(tea leaves)} \\ & x_1, x_2 \geq 0. & \end{aligned}$$

An employee introduces slack variables s_1, s_2, s_3, s_4 and arrives at the following tableau with the basis $\{x_1, x_2, s_3, s_4\}$, which is feasible but has *wrong Row 0 values*, since he has not taken an OR course in his life.

Basis	x_1	x_2	s_1	s_2	s_3	s_4	RHS
Row 0	0	-1	2	1	0	0	0
x_1	1	0	$1/3$	$-1/3$	0	0	20
x_2	0	1	$-1/6$	$2/3$	0	0	10
s_3	0	0	$-1/3$	$1/3$	1	0	10
s_4	0	0	$1/12$	$-1/3$	0	1	15

0 1 a) What is the correct Row 0?

Recalling the revised simplex, Row 0 is given by $-(c_N - (B^{-1}N)^T c_B)^T$. The correct Row 0 is

Basis	x_1	x_2	s_1	s_2	s_3	s_4	RHS
Row 0	0	0	0	0	0	0	0

0 1 b) After the correction of Row 0, is the tableau optimal? Why?

With all Row 0 entries ≥ 0 (here all zero), no negative indicators remain, so the tableau is optimal.

c) The café wants to keep the price of Iced Tea at \$0 and find a price $\$p_1$ for Lemonade. Derive the range of p_1 for which the current basis remains optimal. Is it possible to find an optimal solution that sells Lemonade ($p_1 > 0$) and keeps offering Iced Tea for free?

0
1
2
3
4
5
6

Let $\mathbf{c}_B = (p_1, 0, 0, 0)$ and let B^{-1} be the inverse of the 4×4 basis matrix (whose rows appear under s_1, s_2, s_3, s_4). Then, the Row 0 changes, and the simplex tableau needs some iterations to go back to canonical form for its current basis.

Basis	x_1	x_2	s_1	s_2	s_3	s_4	RHS
Row 0	$-p_1$	0	0	0	0	0	0
Row 0	0	0	$p_1/3$	$-p_1/3$	0	0	$20p_1$

One finds that the reduced costs of the non-basic slacks are

$$c'_{s_1} = p_1/3, \quad c'_{s_2} = -p_1/3.$$

For optimality in a maximization problem, every non-basic reduced cost must satisfy $c'_j \geq 0$. Hence

$$p_1/3 \geq 0 \implies p_1 \geq 0, \quad \text{and} \quad -p_1/3 \geq 0 \implies p_1 \leq 0.$$

Together, these give the unique solution $p_1 = 0$. So, it is not possible to sell Lemonade and keep offering Iced Tea for free.

0 1 2 3 4 5 6

d) Keeping all prices at zero, under what perturbation of the water capacity does the current BFS (basic feasible solution) become *degenerate*? Show which basic variable first hits zero.

If we change the water capacity by Δ, then the simplex tableau is affected as follows:

Basis	x_1	x_2	s_1	s_2	s_3	s_4	RHS
Row 0	0	0	0	0	0	0	0
x_1	1	0	$1/3$	$-1/3$	0	0	$20 + \Delta/3$
x_2	0	1	$-1/6$	$2/3$	0	0	$10 - \Delta/6$
s_3	0	0	$-1/3$	$1/3$	1	0	$10 - \Delta/3$
s_4	0	0	$1/12$	$-1/3$	0	1	$15 + \Delta/12$

Then, the current BFS remains feasible if all RHS values remain non-negative:

$$
\begin{aligned}
20 + \Delta/3 \geq 0 \Rightarrow &\quad \Delta \geq -60 \\
10 - \Delta/6 \geq 0 \Rightarrow &\quad \Delta \leq 60 \\
10 - \Delta/3 \geq 0 \Rightarrow &\quad \Delta \leq 30 \\
15 + \Delta/12 \geq 0 \Rightarrow &\quad \Delta \geq -180
\end{aligned}
$$

Hence, $\Delta \in [-60, 30]$. Then, if the water capacity level changes as much as one of the boundaries of Δ (water capacity is either 40 or 130), the BFS becomes degenerate. In the first case, x_1 hits zero, whereas in the second case, s_3 hits zero.

e) Keeping all prices at zero, introduce a new drink, *Soft Churchill* that uses 3 L water and 1 kg lemon and would sell at \$$p_S$. Write its reduced cost formula using the optimal tableau from part b and state the condition on p_S for the new drink to enter the basis.

0
1
2

Recalling the revised simplex, Row 0 is given by $-(c_N - (B^{-1}N)^T c_B)^T$. For the coefficients $N_S = (3, 0, 1, 0)$ and price p_S, the reduced cost of the new drink can be written as

$$c'_S = -p_S + (B^{-1}N_S)^T 0 = -p_S.$$

The new variable enters the basis if $c'_S < 0$, i.e. $p_S > 0$.

Problem 3 Duality (16 credits)

Consider the following primal linear program (P):

$$
\begin{array}{lll}
\text{Maximize} & 3x_1 + 2x_2 + x_3 & \\
\text{subject to} & x_1 + 2x_2 + x_3 & \leq 10 \\
 & 2x_1 - x_2 + 2x_3 & \geq 4 \\
 & x_1 + x_2 - x_3 & = 3 \\
 & x_1, x_2, x_3 & \geq 0.
\end{array}
$$

0 1 2 3 4 5 6 7

a) Derive the dual linear program (D) corresponding to (P).

The dual problem (D) is:

$$
\begin{array}{lll}
\text{Minimize} & 10y_1 + 4y_2 + 3y_3 & \\
\text{subject to} & y_1 + 2y_2 + y_3 & \geq 3 \\
 & 2y_1 - y_2 + y_3 & \geq 2 \\
 & y_1 + 2y_2 - y_3 & \geq 1 \\
 & y_1 \geq 0 & \\
 & y_2 \leq 0 & \\
 & y_3 \in \mathbb{R} &
\end{array}
$$

0 1 2 3 4

b) State the primal and the dual complementary slackness conditions.

Primal CS conditions:

$$
\begin{aligned}
(x_1 + 2x_2 + x_3 - 10)y_1 &= 0 \\
(2x_1 - x_2 + 2x_3 - 4)y_2 &= 0 \\
(x_1 + x_2 - x_3 - 3)y_3 &= 0
\end{aligned}
$$

Dual CS conditions:

$$
\begin{aligned}
(y_1 + 2y_2 + y_3 - 3)x_1 &= 0 \\
(2y_1 - y_2 + y_3 - 2)x_2 &= 0 \\
(y_1 + 2y_2 - y_3 - 1)x_3 &= 0
\end{aligned}
$$

c) Show that the primal solution $x = (2, 2, 1)^T$ is not optimal.

0
1
2
3
4
5

Assuming that the point x is optimal, there must exist a corresponding feasible point y of the dual problem such that the complementary slackness conditions are satisfied (complementary slackness theorem).
We first check the feasibility of x by inserting it into the primal constraints:

$$\begin{aligned}
&2 + 2 \cdot 2 + 1 = 7 \leq 10\\
&2 \cdot 2 - 2 + 2 \cdot 1 = 4 \geq 4\\
&2 + 2 - 1 = 3\\
&2 \geq 0, 2 \geq 0, 1 \geq 0.
\end{aligned}$$

It follows that x is a primal feasible point. Next, we check if we can find a feasible dual point such that the complementary slackness conditions are satisfied.
Since the first primal constraint is not tight, we must have $y_1 = 0$.
Because $x_2, x_3 \neq 0$, we must have $2y_1 - y_2 + y_3 - 2 = 0$ and $y_1 + 2y_2 - y_3 - 1 = 0$. Since $y_1 = 0$, it follows that $-y_2 + y_3 = 2$ and $2y_2 - y_3 = 1$, i.e., $y_2 = 3$ and $y_3 = 5$. This gives the dual point $y = (0, 3, 5)^T$. We observe that the point cannot be feasible as it must hold $y_2 \leq 0$, but we found $y_2 = 3$. Thus, x cannot be an optimal primal point.

Problem 4 School Planning (22 credits)

Welcome to *Newville* – a city so new it still smells like fresh paint. The city planners are deciding where to build schools. There are $I = \{1, \dots, 15\}$ possible school sites, each with a capacity of $c_i \in \mathbb{R}_{>0}$ and a construction cost of $f_i \in \mathbb{R}_{>0}$ Euros. The city has $J = \{1, \dots, 800\}$ students who need to be assigned to schools. Each student $j \in J$ has a preference score $s_{i,j} \in \mathbb{R}_{>0}$ for each school site $i \in I$, where $s_{i,j} > s_{i',j}$ indicates that student j prefers school site i over i'. Your task is to help the city planners decide there to build schools and how to assign students to schools.

Formulate an integer linear program. Use the binary variable $y_i \in \{0, 1\}$ which is 1 if a school is built at site $i \in I$ and 0 otherwise. Additionally, use the binary variable $x_{i,j} \in \{0, 1\}$ which is 1 if student $j \in J$ is assigned to school $i \in I$ and 0 otherwise. You may introduce additional variables. If you do so, please also state their intuitive meaning in words.

0 1 2

a) Each student must be assigned to exactly one school.

$\sum_{i \in I} x_{i,j} = 1 \ \forall j \in J$

0 1 2

b) The total number of students assigned to each school cannot exceed the school's capacity.

$\sum_{j \in J} x_{i,j} \leq c_i \ \forall i \in I$

0 1 2

c) A student may be assigned to a school only if the school is built.

$x_{i,j} \leq y_i \ \forall i \in I, j \in J$

d) Let the parameter $d_{i,i'} \in \mathbb{R}_{>0}$ denote the distance (in kilometers) between two school sites $i, i' \in I$ with $i \neq i$. The distance between any two built schools must be at least 5 km.

0 1 2 3 4 5

We introduce an auxiliary variable $z_{i,i'} \in \{0, 1\}$: both schools i and i' are open $\implies z_{i,i'} = 1$:

$$1 + z_{i,i'} \geq y_i + y_{i'} \;\forall i, i' \in I, i \neq i'$$

Then we need the constraint:

$$d_{i,i'} z_{i,i'} \geq 5 z_{i,i'} \;\forall i, i' \in I, i \neq i'$$

~~Alternative solution:~~

$$\cancel{M(1 - y_i) + y_i * d_{i,i'} \geq 5y_{i'} \;\forall i, i' \in I, i \neq i}$$

~~Other solutions are also possible.~~

0 1 2 3 4 5 6 7 8

e) If at least one of schools 1 and 2 is assigned more than 100 students, then at least 100 students must be assigned to school 3.

To simplify notation, we define the variable

$$X_i = \sum_{j \in J} x_{i,j} \; \forall i \in \{1, 2, 3\}$$

which indicates the number of students assigned to school $i \in \{1, 2, 3\}$.
Let $t_i \in \{0, 1\}$ be a binary variable which is 1 if more than 100 students are assigned to school $i \in I$ and 0 otherwise.
We introduce an auxiliary variable $q \in \{0, 1\}$: at least one of the schools 1 and 2 has more than 100 students assigned $\iff$ $q = 1$:

$$X_i \leq 100 + Mt_i \; \forall i \in \{1, 2\}$$

$$X_i \geq 101 t_i \; \forall i \in \{1, 2\}$$

$$q \geq \frac{1}{2}(t_1 + t_2)$$

$$q \leq t_1 + t_2$$

Finally, we need to enforce:

$$X_3 \geq 100q$$

Other solutions are also possible.

f) Formulate a single objective function that minimizes total construction costs while maximizing total preference scores of students.

0
1
2
3

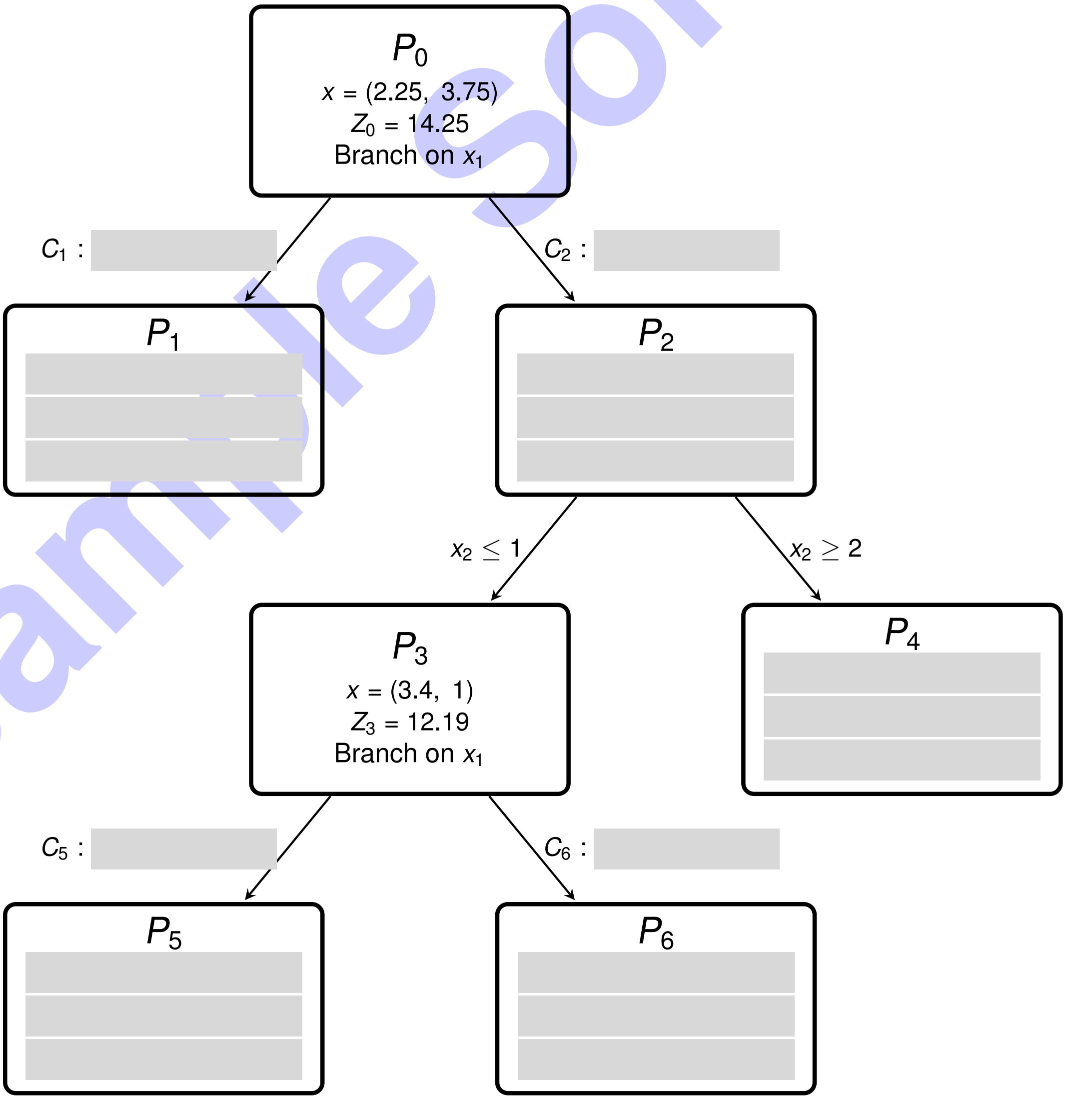
$\min \sum_{i\in I} f_i y_i - \sum_{i\in I} \sum_{j\in J} s_{i,j} x_{i,j}$

Problem 5 Branch and Bound (16 credits)

The great wizard *Gondolf* used the Branch-and-Bound method to solve the following integer program:

$$\begin{aligned} \max\ & 3x_1 + 2x_2 \\ \text{s.t. } & 2.4x_1 + x_2 \leq 9.15 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{N}_0. \end{aligned}$$

To document his journey, Gondolf sketched the branch tree he followed and compiled a table listing all LP relaxations he considered – including some he didn't end up using. In his notation, the subscript in a node name (e.g. P_1) refers to the constraint that was added, and whenever a branching choice was ambiguous, he chose to **branch on the first variable**. He assigned **greater-than-or-equal constraints to right-hand branches** by convention. Unfortunately, Gondolf's clumsy friend *Balbo Buggins* spilled ink on the scroll, obscuring parts of the solution. Can you help recover the missing information and answer the following questions? **Do not write your solution directly on the branch tree diagram.** Please write all answers clearly in the solution box provided below.



Subproblem	Constraints added to original LP	x_1	x_2	Z
P_A	-	2.25	3.75	14.25
P_B	$x_1 \leq 2$	2.00	4.00	14.00
P_C	$x_1 \geq 3$	3.00	1.95	12.90
P_D	$x_2 \leq 3$	2.56	3.00	13.69
P_E	$x_2 \geq 4$	2.00	4.00	14.00
P_F	$x_1 \leq 1, x_2 \geq 4$	1.00	5.00	13.00
P_G	$x_1 \leq 2, x_2 \geq 4$	2.00	4.00	14.00
P_H	$x_1 = 3, x_2 \leq 1$	3.00	1.00	11.00
P_I	$x_1 \geq 2, x_2 \leq 3$	2.56	3.00	13.69
P_J	$x_1 \geq 3, x_2 \leq 1$	3.40	1.00	12.19
P_K	$x_1 \geq 3, x_2 \leq 3$	3.00	1.95	12.90
P_L	$x_1 \geq 3, x_2 \geq 2$	Infeasible		
P_M	$x_1 \geq 4, x_2 \leq 1$	Infeasible		
P_N	$x_1 \geq 4, x_2 \leq 2$	Infeasible		
P_O	$x_1 \geq 4, x_2 \geq 2$	Infeasible		

a) Determine the missing constraints C_1, C_2, C_5, and C_6 (e.g. $x_2 \leq 1$).

0 1 2 3 4

- $C_1 : x_1 \leq 2$
- $C_2 : x_1 \geq 3$
- $C_5 : x_1 = 3$ (also valid $x_1 \leq 3$)
- $C_6 : x_1 \geq 4$

b) Identify which subproblems from the table correspond to the illegible nodes in the tree P_1, P_2, P_4, P_5, P_6 (e.g., $P_0 = P_A$).

0 1 2 3 4 5

- $P_1 = P_B$
- $P_2 = P_C$
- $P_4 = P_L$
- $P_5 = P_H$
- $P_6 = P_M$

c) What are the values of x_1 and x_2 in the optimal integer solution, and what is the corresponding objective value?

0 1

$P_1 : x = (2, 4), Z = 14$

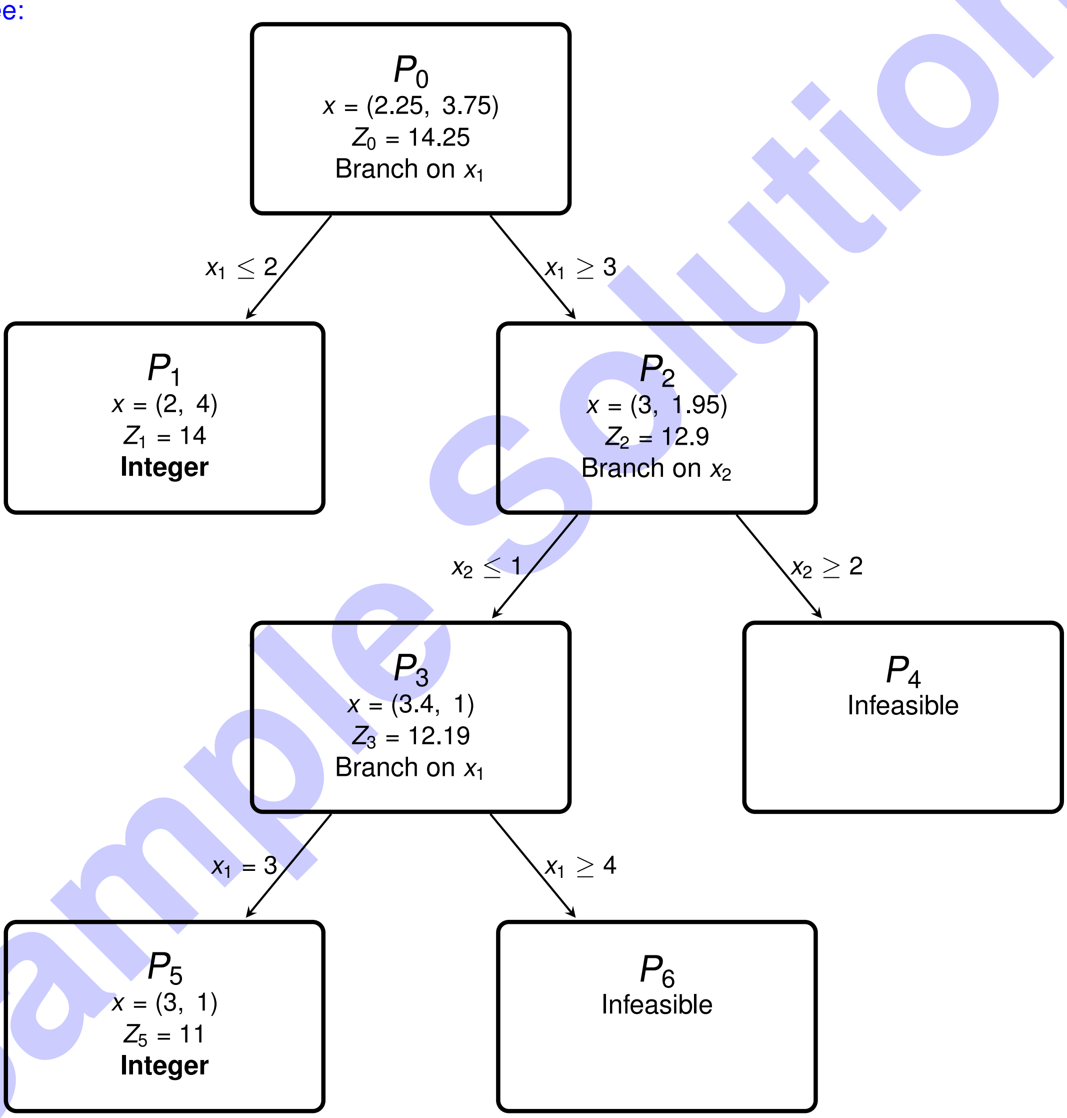
0 1 2 3 4 5 6

d) In which order would Gondolf have explored the nodes, assuming he used the following strategies (and stopped as soon as the optimal solution was confirmed) (e.g. $P_0, P_2, P_3,$ STOP)?

1. FIFO (First-In-First-Out)
2. LIFO (Last-In-First-Out)

1. $P_0, P_1, P_2,$ STOP (early stopping, because P_1 is integer with an OF value of 14 which is larger than the OF value of P_2 and with it larger than any child of P_2)
2. $P_0, P_2, P_4, P_3, P_6, P_5, P_1$

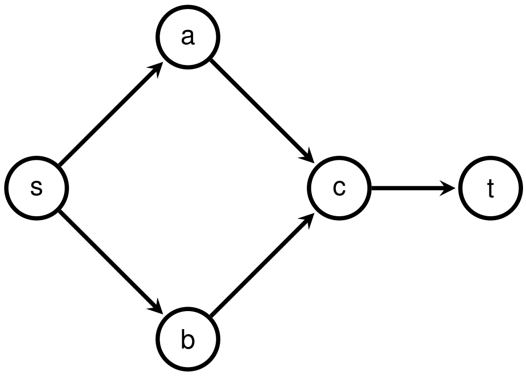
Full tree:



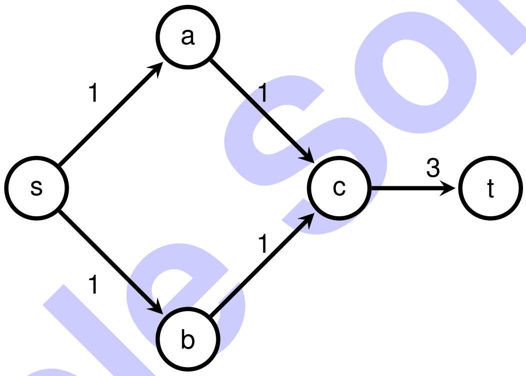
Problem 6 Maximum flow and minimum cut counterexamples (14 credits)

Let $(G = (V, E), s, t, c))$ be a network with source s, sink t, and integral capacities $c(e) \in \mathbb{N}$ for all edges $e \in E$.
Consider the following statements below. Use the given graphs to find a suitable counterexample for each one of them. Clearly describe why the given counterexample refutes the statement.

a) If all capacities are odd then there is a maximal $(s - t)$-flow f such that $f(e)$ is odd for all $e \in E$.

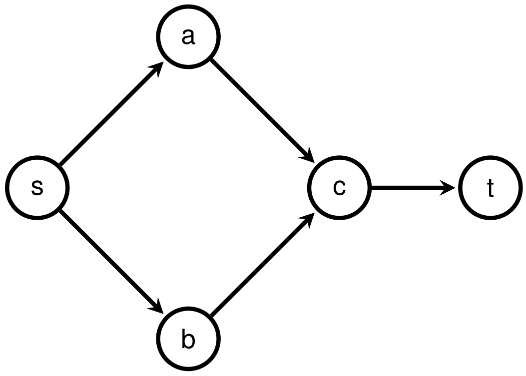


Consider the following graph:

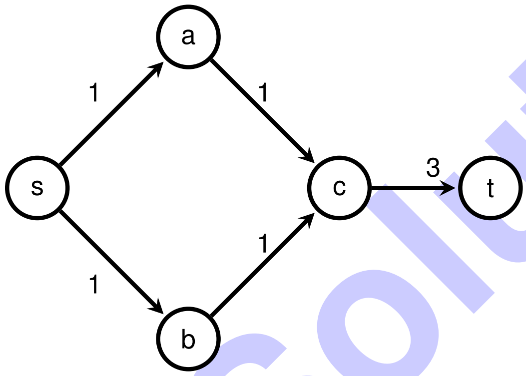


All capacities are odd. The maximum $s - t$-flow has a value of 2, which flows over $e := (c, t)$. The flow $f(e) = 2$ is an even number. Therefore, even though all capacities are odd integer numbers, the maximum flow has a flow over edge e which is an even number, showing that the statement is false in general.

b) Adding a number $\lambda \in \mathbb{N}$ to all capacities $c(e)$ does not change the minimal cuts.



Consider the following graph:



A minimal cut of value 2 is given by $(\{s\}, \{a, b, c, t\})$. After adding $\lambda = 2$ to all capacities, the minimal value of 5 is achieved by the cut $(\{t\}, \{s, a, b, c\})$, while the value of the initial cut $(\{s\}, \{a, b, c, t\})$ is 6. Therefore, the set of minimal cuts has changed, which shows that the statement is false.

Problem 7 Minimal Circle Enclosure (22 credits)

Matilda has been hired by Amazon to determine the optimal location for a new warehouse that will serve n distribution centers. Let the positions of the distribution centers be given by points $p_i = (x_i, y_i) \in \mathbb{R}^2$ for $i = 1, \dots, n$. Assume that at least two distribution centers have distinct locations, i.e., there exist j, k such that $p_j \neq p_k$.

Help Matilda find the optimal warehouse location $M = (c_1, c_2) \in \mathbb{R}^2$ that minimizes the maximum distance to any distribution center. In other words, find the center M and radius $r \geq 0$ of the smallest circle such that all points p_i lie inside or on the boundary of this circle.

a) State the definition of a convex function.

0 1

Let $x, y \in \mathbb{R}^n$, $\lambda \in [0, 1]$. $f : \mathbb{R}^n \to \mathbb{R}$ is called convex if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

b) Prove that any norm $\|\cdot\|$ on $\mathbb{R}^d$ is a convex function.

Hint: A norm satisfies the following properties for all $x, y \in \mathbb{R}^d$, and $\alpha \in \mathbb{R}$:

0 1 2

- Positive definiteness: $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$.
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$.
- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$.

$$||\lambda x + (1 - \lambda)y|| \leq ||\lambda x|| + ||(1 - \lambda)y|| = |\lambda|||x|| + |(1 - \lambda)|||y||$$

The inequality follows by the Triangle inequality. The last equality follows by Homogeneity.

c) Formulate the problem of finding the smallest enclosing circle as a convex optimization problem. Specifically, express it in the form:

0 1 2 3 4

$$\begin{aligned} \min_{r,c_1,c_2} \quad & f(r, c_1, c_2) \\ \text{s.t.} \quad & g_i(r, c_1, c_2) \leq 0 \quad \text{for all } i = 1, \dots, n \\ & r \geq 0, \quad c_1, c_2 \in \mathbb{R} \end{aligned}$$

Show that the objective function f and the constraint functions g_i are convex.

Hint: Use the Euclidean norm $\|\cdot\|_2$.

$$\begin{aligned} &\min_{r,c_1,c_2} r \\ &\text{s.t. } ||M - p_i||_2 - r \leq 0 \ \forall i \\ &\quad c_i \in \mathbb{R}, r \geq 0 \end{aligned}$$

$f(r, c_1, c_2) = r$ is linear and thus convex.
$g_i(r, c_1, c_2,) = ||M - p_i||_2 - r$ is by b) the sum of two convex functions and thus convex.

0
1
2
3
4

d) Show that Slater's condition holds for this problem. That is, find a specific point (r, c_1, c_2), expressed in terms of the given $\{(x_i, y_i)\}$, such that all inequality constraints are satisfied strictly.

$$\bar{c}_1 := \frac{1}{n}\sum_{i=1}^{n} x_i$$
$$\bar{c}_2 := \frac{1}{n}\sum_{i=1}^{n} y_i$$
$$\bar{M} := (\bar{c}_1, \bar{c}_2)$$
$$\bar{r} := \max_i ||\bar{M} - p_i||_2 + 1$$

Then it holds that $g_i \leq -1 < 0$.

0
1
2
3
4

e) Given the Lagrangian function:

$$L = f + \sum_{i=1}^{n} \lambda_i g_i - \mu r, \quad \text{with } \lambda_i, \mu \geq 0$$

State the KKT conditions.

$$\begin{aligned}
\text{Stationarity: } \nabla L &= 0 \\
\text{Primal feasibility: } g_i &\leq 0, n, \quad -r \leq 0 \\
\text{Complementary slackness: } \lambda_i g_i &= 0, \mu r = 0 \\
\text{Dual feasibility: } \lambda_i &\geq 0, n, \quad \mu \geq 0
\end{aligned}$$

f) Use the **(KKT) conditions** to show that at the optimal solution (c_1^*, c_2^*), the following must hold:

0 1 2 3 4 5 6

$$1 = \sum_{i=1}^{n} \lambda_i$$
$$c_1^* = \sum_{i=1}^{n} \lambda_i x_i$$
$$c_2^* = \sum_{i=1}^{n} \lambda_i y_i$$

That is, the optimal warehouse location is a convex combination of the distribution center locations.

Hint: Use that there are at least two distinct distribution centers.

Since there exists two different distribution centres, it follows $r > 0$. By complementary slackness we have $\mu r = 0$ thus $\mu = 0$.

$$\frac{d}{dr}L = 1 - \sum_{i=1}^{n} \lambda_i = 0 \Leftrightarrow 1 = \sum_{i=1}^{n} \lambda_i$$

$$\frac{d}{dc_1}L = \sum_{i=1}^{n} \lambda_i \frac{c_1 - x_i}{||p_i - M||_2} = 0$$

By complementary slackness it must hold that either $\lambda_i = 0$ or $||p_i - M||_2 = r$. Thus it follows

$$\sum_{i=1}^{n} \lambda_i \frac{c_1 - x_i}{r} = 0 \Leftrightarrow \sum_{i=1}^{n} \lambda_i x_i = \sum_{i=1}^{n} \lambda_i c_1 = c_1 \sum_{i=1}^{n} \lambda_i = c_1$$

The result for c_2 follows analogously.

g) Suppose $\lambda_i > 0$ for some index i. What does this imply about the corresponding point p_i in relation to the minimal enclosing circle?

0 1

p_i is on the boundary of the minimal enclosing circle.

Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.